$$\begin{pmatrix} p_1(t+1) \\ p_2(t+1) \end{pmatrix} = \begin{pmatrix} 1-\varepsilon & 0 \\ \varepsilon & 1 \end{pmatrix} \begin{pmatrix} p_1(t) \\ p_2(t) \end{pmatrix}$$

$$p_1(t+1) = p_1(t)(1-\varepsilon)$$

$$p_2(t+1) = \varepsilon(p_1(t)) + p_2(t)$$

$$p_1(t) = \frac{p_1(t+1)}{(1-\varepsilon)}$$

$$p_2(t) = p_2(t+1) - \varepsilon(p_1(t))$$

$$p_1(t) + p_2(t) = \frac{p_1(t+1)}{(1-\varepsilon)} - \varepsilon(p_1(t)) + p_2(t+1)$$

$$= p_1(t) - \varepsilon(p_1(t) + \varepsilon$$

$$(1-\varepsilon)p_1(t-1) + \varepsilon p_1(t-1) + p_2(t-1)$$

$$p_1(t-1) - \varepsilon(p_1(t-1)) + \varepsilon p_1(t-1)$$
$$+ p_2(t-1)$$

$$\begin{pmatrix} p_1(t+1) \\ p_2(t+1) \end{pmatrix} = \begin{pmatrix} 1-\varepsilon & 0 \\ \varepsilon & 1 \end{pmatrix} \begin{pmatrix} p_1(t \\ p_2(t \end{pmatrix}$$

$$p_1(t+1) = (1-\varepsilon)p_1(t) = p_1(t) - \cancel{\varepsilon p_1(t)}$$

$$p_2(t+1) = \varepsilon \cancel{p_1(t)} + p_2(t)$$

$$p_1(t+1) + p_2(t+1) = p_1(t) + p_2(t)$$

$$p_1(t) + p_2(t) = p_1(t-1) + p_2(t-1)$$

$$p_1(t) = p_1(t-t) + p_2(t-t)$$

$$= p_1(0) + p_2(0)$$

$$= C$$

$$M\vec{v_1} = \lambda_1 \vec{v_1}$$

$$\begin{bmatrix} 1-\varepsilon & 0 \\ \varepsilon & 1 \end{bmatrix} \vec{v_1} = \lambda_1 \vec{v_1}$$

$$v_1 = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

$$(1-\varepsilon)x_1 + 0 + \varepsilon x_1 + y_1 = \lambda_1(x_1 + y_1)$$

$$x_1 - \cancel{\varepsilon x_1 + \varepsilon x_1} + y$$

$$\cancel{x_1 + y_1} = \lambda_1 (x_1 \cancel{+ y_1})$$

$$\lambda_1 = 1$$

$$\boxed{\det(M - \lambda I) = 0}$$

$$P_r(t = 0) = \frac{1}{2}$$

$$\det(M - \lambda I)$$